Day 4 Hand On Subnetting

Task 1:- Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

- Original CIDR: `/24`
- Number of subnets: 4 = \(2^2\), so borrow 2 bits.
- New CIDR: `/26`
- Subnet mask: 255.255.255.192
- Addresses per subnet: 64 (62 usable host addresses)

Subnetting Table

Subnet	Network Address	Subnet Mask	Broadcast Address
1	192.168.10.0/26	255.255.255.192	192.168.10.63
2	192.168.10.64/26	255.255.255.192	192.168.10.127
3	192.168.10.128/26	255.255.255.192	192.168.10.191
4	192.168.10.192/26	255.255.255.192	192.168.10.255

Valid Host Address Ranges

Subnet	First Usable Host	Last Usable Host
1	192.168.10.1	192.168.10.62
2	192.168.10.65	192.168.10.126
3	192.168.10.129	192.168.10.190
4	192.168.10.193	192.168.10.254

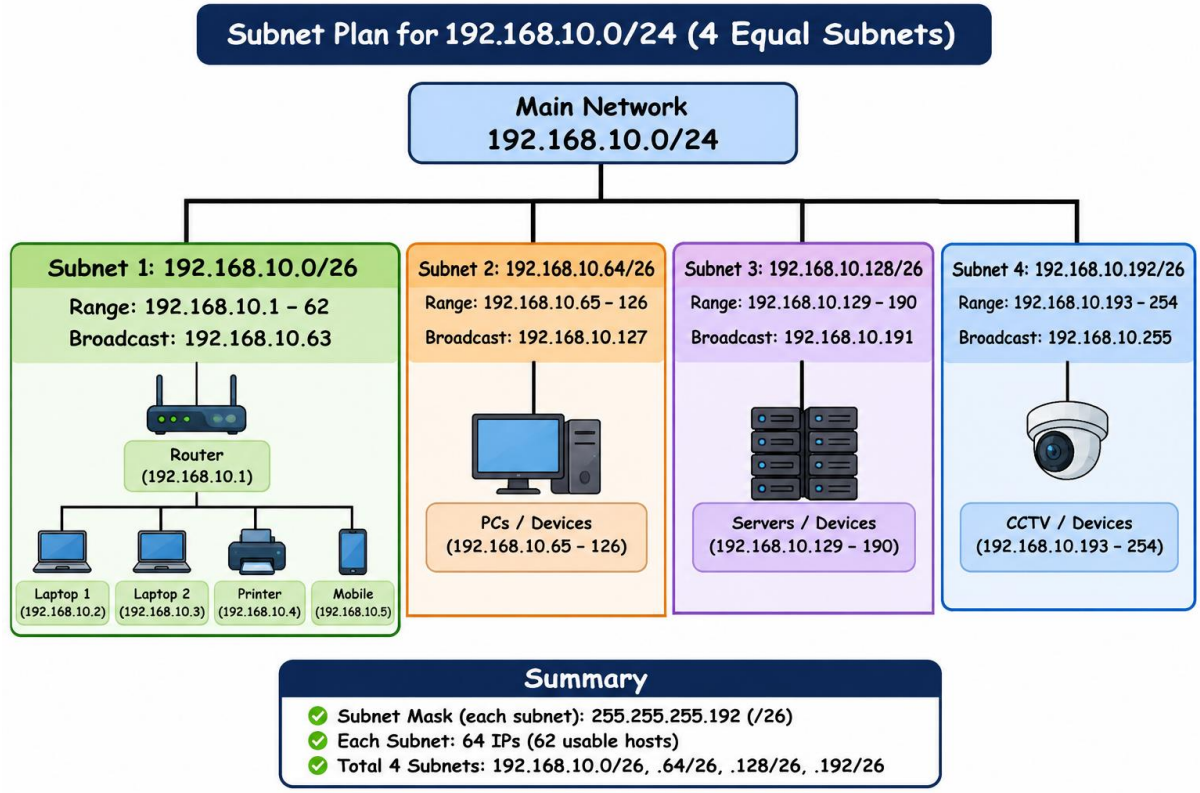
Task 2:- For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device. Make sure not to use the network or broadcast address for any device.

- Subnet Mask: 255.255.255.192
- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63
- Valid Host Range: 192.168.10.1 – 192.168.10.62

IP Address Assignment for 5 Devices

Device	Assigned IP Address	Subnet Mask
Router	192.168.10.1	255.255.255.192
Laptop 1	192.168.10.2	255.255.255.192
Laptop 2	192.168.10.3	255.255.255.192
Printer	192.168.10.4	255.255.255.192
Mobile	192.168.10.5	255.255.255.192

Task 3:- Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.



Task 4:- Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Step-by-Step Subnetting Calculation and IP Assignment

Step 1: Understand the Given Network

The given network address is 192.168.10.0/24.

- IP address: 192.168.10.0
- Original subnet mask: 255.255.255.0
- CIDR notation: /24
- Required number of subnets: 4

The /24 means that 24 bits are used for the network portion, and 8 bits are available for hosts.

Step 2: Calculate the New Subnet Mask

We need 4 equal subnets.

The formula for the number of subnets is:

$$\text{Number of subnets}=2^n$$

Here, n is the number of bits borrowed from the host portion.

$$2^2=4$$

Therefore, we borrow 2 bits from the host portion.

New CIDR prefix:

$$/24+2=/26$$

The new subnet mask is 255.255.255.192 (/26).

Step 3: Calculate the Number of Hosts

There are 32 bits in an IPv4 address. After borrowing 2 bits, the remaining host bits are:

$$32-26=6$$

Total IP addresses per subnet:

$$2^6=64$$

Usable host addresses:

$$64-2=62$$

We subtract 2 because each subnet reserves one address for the network and one for the broadcast.

Therefore, each subnet has 64 total IP addresses and 62 usable host addresses.

Step 4: Calculate the Subnet Ranges

The subnet block size is:

\[256-192=64 \]

So, the subnet network addresses increase by 64 in the last octet: 0, 64, 128, and 192.

Subnet	Network Address	First Usable IP	Last Usable IP	Broadcast Address
1	192.168.10.0/26	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64/26	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128/26	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192/26	192.168.10.193	192.168.10.254	192.168.10.255

The subnet mask for all four subnets is 255.255.255.192.

Step 5: Assign IP Addresses to Devices

I selected Subnet 1: 192.168.10.0/26 for my five devices.

Device	Assigned IP Address	Subnet Mask
Router	192.168.10.1	255.255.255.192
Laptop 1	192.168.10.2	255.255.255.192
Laptop 2	192.168.10.3	255.255.255.192
Printer	192.168.10.4	255.255.255.192
Mobile	192.168.10.5	255.255.255.192

I assigned the router the first usable IP address and gave each remaining device a different IP address from the same subnet.

Step 6: Verify the IP Assignments

I checked that:

1. All five devices have unique IP addresses.
2. All addresses are within the valid host range of 192.168.10.1 to 192.168.10.62.
3. The network address 192.168.10.0 is not assigned to a device.
4. The broadcast address 192.168.10.63 is not assigned to a device.
5. All devices use the same subnet mask.